

SMART INDIA HACKATHON 2025



- **Problem Statement ID :** 25106
- **Problem Statement Title:** Internship/Industrial Training with Placement Opportunity
- **Theme:** Smart Education
- **PS Category-** Software
- **Team ID-**
- **Team Name -** AltF4



Employability OS (e-OS)

What is e-OS: It is an AI-powered platform that replaces the chaotic maze of emails, WhatsApp, and spreadsheets with a single, intelligent system for placements. Our platform provides a one stop solution for Internships and Placement.

Current System Problems:

- Chaotic & Inefficient
- Scattered communication

Innovation:

- Recommendation Engine
- Real-time Dashboard
- Role-Based Access

Technical Approach

Technologies Used (Tech Stack)

React.js, Node.js (Express.js), MongoDB, ML Libraries (Scikit-learn, TensorFlow etc.)

01

We'll define and create the MongoDB schemas and Using Node.js & Express.js, we'll build the essential REST APIs

02

We'll build authentication system using JSON Web Tokens (JWT) for secure registration and login system

03

We'll develop a Recommendation Engine using Scikit Learn and TensorFlow

04

We'll build a content-based filtering model using Scikit-learn.

05

Using React.js we'll build frontend and integrate all the frontend components with our backend APIs

Feasibility And Viability

Technical Feasibility

- Our phased development roadmap ensures that a core, functional MVP (Minimum Viable Product) is achievable within the hackathon timeframe.
- The chosen tech stack (React, Node.js, Scikit Learn) is mature, well-documented, and our team has the required expertise to implement it effectively.

Economic Viability

- The platform can operate on a low-cost subscription (SaaS) model for institutions, making it affordable and accessible.
- It's highly scalable, capable of expanding to universities and colleges across the country with minimal additional infrastructure cost.
- Streamlines processes and reduces manual effort, save student and recruiter's time to save valuable time in administration and operations.

Impact And Benefits

Direct Impact on Stakeholders

- **For Students:** Transforms a stressful job hunt into a clear, guided career opportunities.
- **For the Placement Cell:** Eliminates over 90% of manual, repetitive tasks, freeing them to focus on strategic industry partnerships and student mentoring.
- **For the College:** Provides data-driven insights which boosts placement statistics

Broader National Benefits

- **Economic Growth:** By helping companies hire the right talent faster, we reduce recruitment costs and contribute to a more productive workforce.
- **Skill India Initiative:** Our platform directly supports the national goal of skilling the youth by identifying in-demand skills and helping students acquire them.
- **Bridging the Skill Gap:** We provide a crucial link between academic knowledge and industry requirements, ensuring graduates are truly job-ready.

Research And References

Details / Links of the reference and research work:

[1] https://www.irjmets.com/uploadedfiles/paper/issue_3_march_2025/69924/final/fin_irjmets1742653988.pdf

[2] <https://www.jetir.org/papers/JETIR2504302.pdf>

[3] <https://foundryjournal.net/wp-content/uploads/2025/08/32.FJ25C734.pdf>

[4] <https://www.irjet.net/archives/V11/i1/IRJET-V11I104.pdf>

[5] <https://www.cloudeagle.ai/blogs/role-based-access-control>

[6] https://ijaem.net/issue_dcp/Recommendation%20system%20for%20student%20placement.pdf